

Part II — Motion and Forces · Chapter 13

Circular Motion and Gravitation

Understanding why curved motion requires inward force, and how gravity governs planets, satellites, and orbital mechanics

A car rounds a curve at 60 km/h. A satellite orbits 400 km above Earth. A stone swings on a string. In each case, something is changing the direction of motion without necessarily changing the speed. Newton's first law tells us this requires a force. But which force, and in what direction? This chapter shows that curved motion always requires a net force directed inward — toward the center of curvature — and that gravity is the most important source of that force in nature.

13.1 Why Circular Motion Is Different from Linear Motion

In linear motion, acceleration changes the magnitude of velocity. In circular motion at constant speed, acceleration changes the direction of velocity while the magnitude stays constant. Both are accelerations — both require a net force by Newton's second law — but they feel very different and require different analysis.

A key insight: an object moving in a circle at constant speed is always accelerating, even though its speedometer would read constant. The velocity vector is always tangent to the circle, and since the direction of a tangent changes continuously as the object moves around, the velocity is constantly changing. Any change in velocity is acceleration.

The acceleration required to maintain circular motion points toward the center of the circle at every instant. This is called centripetal acceleration ('centripetal' comes from the Latin for 'center-seeking'). The corresponding force is the centripetal force.

Key Point: Centripetal force is not a new type of force. It is simply the name for whatever force (or net force) is directed toward the center of curvature and provides the centripetal acceleration. Gravity, tension, friction, and normal force can all serve as centripetal force in different situations.

13.2 Velocity in Circular Motion

For an object moving in a circle of radius r with constant speed v , the velocity vector is always tangent to the circle — perpendicular to the radius at every point. The speed v

is the magnitude of this velocity vector and remains constant for uniform circular motion.

The period T is the time to complete one full circle. The speed is the circumference divided by the period:

$$v = 2\pi r / T$$

v = speed (m/s); r = radius (m); T = period (s). For constant speed circular motion only.

The frequency f is the number of complete circles per second (in Hz). Frequency and period are reciprocals: $f = 1/T$.

Example — Circular speed A satellite orbits Earth at radius $r = 7.0 \times 10^6$ m with period $T = 5820$ s. Speed $= 2\pi \times 7.0 \times 10^6 / 5820 = 7.56 \times 10^3$ m/s $= 7.56$ km/s.

13.3 Centripetal Acceleration

Even at constant speed, an object in circular motion has acceleration because its velocity direction changes. This centripetal acceleration always points toward the center of the circle.

$$a_c = v^2 / r$$

Centripetal acceleration in m/s^2 ; v in m/s ; r in radius of circular path in m .

An equivalent expression using period T or frequency f :

$$a_c = 4\pi^2 r / T^2 = 4\pi^2 r f^2$$

Centripetal acceleration increases with speed squared and decreases with increasing radius. Double the speed $\rightarrow$ four times the centripetal acceleration. Double the radius $\rightarrow$ half the centripetal acceleration.

Example — Centripetal acceleration A ball on a 0.80 m string moves at 3.0 m/s. $a_c = (3.0)^2 / 0.80 = 9.0 / 0.80 = 11.25$ m/s^2 toward the center.

Speed	Radius	Centripetal Acceleration
v	r	v^2/r
$2v$	r	$4v^2/r$ (quadrupled)
v	$2r$	$v^2/(2r)$ (halved)
$2v$	$2r$	$4v^2/(2r) = 2v^2/r$ (doubled)

Table 13.1 How centripetal acceleration scales with speed and radius.

13.4 Centripetal Force

By Newton's second law, the centripetal acceleration requires a net force directed toward the center of the circle. This is the centripetal force:

$$\mathbf{F}_c = m \mathbf{v}^2 / r$$

Centripetal force in N; m in kg; v in m/s; r in m. Always directed toward the center.

This equation is not a new force law; it is Newton's second law applied to circular motion. Whatever real force provides the centripetal acceleration — gravity, tension, friction, normal force, or any combination — its inward component must equal mv^2/r for the object to maintain circular motion.

The Rule: To solve any circular motion problem: (1) Identify what force (or component of force) points toward the center. (2) Set that force equal to mv^2/r . (3) Solve for the unknown.

13.5 Sources of Centripetal Force

In different physical situations, different forces supply the centripetal acceleration. Recognizing the source of centripetal force is the first step in any circular motion problem.

Situation	Source of Centripetal Force	Direction toward Center
Ball on a string (horizontal circle)	Tension in string	Along string, toward hand
Car rounding a flat curve	Friction from road on tires	Horizontal, toward center of curve
Car on banked curve	Component of normal force	Horizontal inward component of N
Satellite in orbit	Gravity	Toward Earth's center
Ball in a vertical loop (bottom)	Normal force minus weight	Net upward = toward center
Ball in a vertical loop (top)	Weight plus normal force	Net downward = toward center
Planet orbiting Sun	Gravitational attraction	Toward Sun

Table 13.2 Sources of centripetal force in common circular motion situations.

Worked Example 13.1 — Ball on a String (Horizontal Plane)

A 0.25 kg ball swings on a 0.60 m string in a horizontal circle at 4.0 m/s. Find the tension.

Step 1: Centripetal force needed: $F_c = mv^2/r = 0.25 \times (4.0)^2 / 0.60 = 0.25 \times 16.0 / 0.60 = 6.67 \text{ N}$

Step 2: Source of centripetal force: Tension (horizontal component, since string is horizontal)

Step 3: Set equal: $T = 6.67 \text{ N}$

Worked Example 13.2 — Car Rounding a Flat Curve

A 1200 kg car rounds a flat curve of radius 80.0 m at 20.0 m/s. Find the friction force needed and check if it is possible ($\mu_s = 0.50$).

Step 1: Centripetal force needed: $F_c = mv^2/r = 1200 \times 400 / 80.0 = 6000 \text{ N}$

Step 2: Source: Static friction (the car is not sliding sideways if it rounds the curve)

Step 3: Maximum available friction: $f_{s,\max} = \mu_s N = 0.50 \times 1200 \times 9.81 = 5886 \text{ N}$

Step 4: Compare: $6000 \text{ N} > 5886 \text{ N}$ — the car cannot make this curve at 20 m/s with $\mu_s = 0.50$.

13.6 Banking and Circular Motion Applications

A banked curve is tilted at an angle θ inward so that the horizontal component of the normal force provides centripetal acceleration, reducing or eliminating the need for friction. At the ideal banking angle for a given speed, no friction is needed at all.

$$\tan\theta = v^2 / (rg)$$

Ideal banking angle θ for speed v on a curve of radius r ; valid for the speed where no friction is required.

Example — Banking angle A highway curve of radius 120 m is to be banked so cars at 25 m/s need no friction. $\tan\theta = (25)^2 / (120 \times 9.81) = 625 / 1177 = 0.531 \rightarrow \theta = 28.0^\circ$.

Vertical circular motion introduces an additional complication: the gravitational force changes its contribution to centripetal force depending on position in the loop. At the bottom of a loop, the normal force must exceed weight to provide centripetal acceleration upward. At the top of a loop, weight and normal force both point toward the center; the minimum speed occurs when $N = 0$.

Worked Example 13.3 — Minimum Speed at Top of a Loop

A roller coaster loop has radius 12.0 m. What is the minimum speed at the top to maintain contact?

Step 1: At the top, centripetal direction is downward (toward center): $mg + N = mv^2/r$

Step 2: At minimum speed, $N = 0$: $mg = mv_{\min}^2/r \rightarrow v_{\min} = \sqrt{gr} = \sqrt{(9.81 \times 12.0)} = \sqrt{117.7} = 10.9 \text{ m/s}$

13.7 Universal Gravitation

Gravity is universal: every object in the universe with mass attracts every other object with mass. This was Newton's great insight, published in the Principia in 1687: the same force that causes an apple to fall causes the Moon to orbit Earth, and the same force governs planets orbiting the Sun.

Before Newton, gravity and the celestial motions of planets were considered entirely separate phenomena. Newton unified them with a single mathematical law, showed that the inverse-square relationship produced Kepler's elliptical orbits, and demonstrated that the force causing free fall on Earth was exactly sufficient to keep the Moon in its orbit. This unification was one of the greatest intellectual achievements in scientific history.

Historical Note: Newton reportedly said that the idea of universal gravitation came to him when he saw an apple fall in his garden in 1666, during a forced retreat from Cambridge due to plague. Whether the story is literally true is disputed, but the timing — his 'annus mirabilis' — is well documented. He did not publish his results until 1687, spending years verifying the quantitative predictions.

13.8 Newton's Law of Gravitation

The gravitational force between two masses is attractive, acts along the line joining them, and has magnitude:

$$F_g = G m_1 m_2 / r^2$$

F_g in N; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ (universal gravitational constant); m_1, m_2 in kg; r = distance between centers in m.

Key features of the law: the force is always attractive (there is no gravitational repulsion). It is proportional to each mass separately: double m_1 and the force doubles. It is inversely proportional to the square of the separation: double r and the force drops to one-quarter.

Change

Effect on F_g

Double m_1	$F_g \times 2$
Double both masses	$F_g \times 4$
Double the distance r	$F_g \div 4$ (inverse square)
Triple the distance r	$F_g \div 9$
Halve the distance r	$F_g \times 4$

Table 13.3 *How gravitational force scales with mass and distance.*

Worked Example 13.4 — Force Between Two Objects

Find the gravitational force between two 70.0 kg people separated by 1.0 m.

Step 1: Apply law: $F_g = G m_1 m_2 / r^2 = 6.674 \times 10^{-11} \times 70.0 \times 70.0 / (1.0)^2$

Step 2: Calculate: $= 6.674 \times 10^{-11} \times 4900 = 3.27 \times 10^{-7} \text{ N}$

Step 3: Interpretation: About 0.33 micronewtons — utterly negligible. Gravity between everyday objects is undetectable without extremely sensitive instruments.

13.9 Gravitational Fields

Rather than thinking of gravity as an action-at-a-distance force, physicists often describe it using the concept of a gravitational field. The gravitational field $\vec{g}$ at any point in space is defined as the gravitational force per unit mass that would act on a small test mass placed there:

$$\vec{g} = \vec{F}_g / m = -G M / r^2 \quad (\text{pointing toward } M)$$

Gravitational field strength at distance r from mass M ; same units as acceleration: m/s^2 .

Near Earth's surface, the gravitational field is approximately uniform with magnitude $g = 9.81 \text{ m/s}^2$, directed downward. This is the g used throughout Chapters 6–12. As altitude increases, g decreases:

$$g(r) = G M_{\text{Earth}} / r^2$$

Example — g at altitude At 400 km above Earth's surface: $r = 6.37 \times 10^6 + 4.0 \times 10^5 = 6.77 \times 10^6 \text{ m}$. $g = 6.674 \times 10^{-11} \times 5.97 \times 10^{24} / (6.77 \times 10^6)^2 = 8.69 \text{ m/s}^2$. Gravity at the ISS altitude is only about 11% less than at the surface.

13.10 Weight vs. Mass Revisited

We can now revisit the weight-mass distinction with greater precision. The weight of an object is the gravitational force on it: $W = mg$, where g is the local gravitational field strength. Mass m is constant; weight varies with location because g varies.

An astronaut on the ISS (at 400 km altitude) experiences $g \approx 8.69 \text{ m/s}^2$. Their weight is $W = m \times 8.69 \text{ N}$ — not zero! The astronaut appears weightless because the station and the astronaut are in the same free-fall orbit. Both accelerate toward Earth at the same rate, so there is no contact force between them — the normal force is zero, giving the sensation of weightlessness.

Key Point: Astronauts in orbit are not in zero gravity. They are in continuous free fall. The sensation of 'weightlessness' is the absence of a normal force (support force), not the absence of gravity. The gravitational force on the ISS astronauts is about 89% of their surface weight.

13.11 Orbital Motion

An orbit is a continuous free fall around a body. The satellite falls toward Earth due to gravity, but moves so fast horizontally that Earth's surface curves away at the same rate the satellite falls. The result is a circular (or elliptical) path that never intersects the surface.

For a circular orbit, the gravitational force provides exactly the centripetal force needed:

$$G M m / r^2 = m v^2 / r$$

Solving for the orbital speed:

$$v_{\text{orbit}} = \sqrt{(G M / r)}$$

Orbital speed depends only on the mass of the central body M and the orbital radius r — not on the satellite's mass m .

This is a profound result: all objects at the same orbital radius orbit at the same speed, regardless of mass. A space station and an astronaut floating outside it orbit together because they have the same r and the same v_{orbit} .

Worked Example 13.5 — Orbital Speed of the ISS

The ISS orbits at $r = 6.77 \times 10^6 \text{ m}$. Find its orbital speed.

Step 1: Apply formula: $v = \sqrt{(G M_{\text{Earth}} / r)} = \sqrt{(6.674 \times 10^{-11} \times 5.97 \times 10^{24} / 6.77 \times 10^6)}$

Step 2: Numerator: $6.674 \times 10^{-11} \times 5.97 \times 10^{24} = 3.984 \times 10^{14}$

Step 3: Divide by r : $3.984 \times 10^{14} / 6.77 \times 10^6 = 5.885 \times 10^7$

Step 4: Take root: $v = \sqrt{(5.885 \times 10^7)} = 7672 \text{ m/s} \approx 7.67 \text{ km/s}$

13.12 Satellites and Planetary Motion

The orbital period T (time for one complete orbit) is related to orbital radius r by combining $v = 2\pi r/T$ with $v = \sqrt{(GM/r)}$:

$$T^2 = (4\pi^2 / G M) r^3$$

Kepler's Third Law: the square of the period is proportional to the cube of the orbital radius.

This relationship, first discovered empirically by Johannes Kepler in 1619 from astronomical observations, was later derived theoretically by Newton from his gravitational law. It is one of the great confirmations that Newton's mechanics and gravity are correct.

Object	Orbital Radius (m)	Period	Orbital Speed (m/s)
Moon	3.84×10^8 m	27.3 days	1020
ISS	6.77×10^6 m	92.7 min	7670
GPS satellite	2.66×10^7 m	12 h	3870
Geostationary sat.	4.22×10^7 m	24 h	3070
Earth around Sun	1.50×10^{11} m	365.25 days	29,800

Table 13.4 *Orbital parameters for selected objects. Note: orbital speed decreases with increasing radius.*

13.13 Escape Velocity and Orbital Concepts

Escape velocity is the minimum speed required for an object to escape a planet's gravitational pull entirely — to reach infinite distance without further propulsion. It is found by setting the object's kinetic energy equal to the work needed to escape the gravitational well:

$$v_{\text{escape}} = \sqrt{(2 G M / r)}$$

Escape velocity from a body of mass M at distance r from its center.

Note that $v_{\text{escape}} = \sqrt{2} \times v_{\text{orbit}}$: escape velocity is $\sqrt{2} \approx 1.41$ times the circular orbital speed at the same radius. For Earth's surface: $v_{\text{escape}} = \sqrt{(2 \times 6.674 \times 10^{-11} \times 5.97 \times 10^{24} / 6.37 \times 10^6)} \approx 11.2$ km/s.

Body	Surface g (m/s ²)	Escape Velocity (km/s)
Earth	9.81	11.2
Moon	1.62	2.38
Mars	3.71	5.03

Jupiter	24.8	59.5
Sun	274	618
Black hole	Extreme	$> c$ (light cannot escape)

Table 13.5 *Surface gravity and escape velocities for selected bodies.*

Geosynchronous orbit is the radius at which the orbital period equals Earth's rotation period (24 hours). A satellite in geosynchronous orbit above the equator appears stationary from the ground — this is the basis for communication and weather satellites. Geosynchronous radius $\approx 42,200$ km from Earth's center.

Bridge Note *In Volume I of the advanced series, planetary motion is derived from the Lagrangian in a central force potential. Kepler's first law (elliptical orbits) requires conserved angular momentum, which emerges naturally from the central-force symmetry via Noether's theorem.*

13.14 Common Mistakes in Circular Motion Analysis

Mistake 1: Inventing a 'centrifugal force' pointing outward

In an inertial reference frame, there is no outward centrifugal force. The sensation of being pushed outward in a turning car is inertia — your body tends to continue in a straight line while the car turns. Centrifugal force is a fictitious force that appears only in rotating (non-inertial) reference frames. In all problems in this book, work in an inertial frame and include only real inward forces on the FBD.

Common Mistake: Never add a centrifugal force pointing outward to a free-body diagram for circular motion. The centripetal force IS the net inward force. If you draw something outward, you are double-counting or inventing a fictitious force.

Mistake 2: Setting centripetal force equal to a single force when multiple forces act

The centripetal force is the net inward force. Write Newton's second law in the radial (inward) direction and sum all inward force components. Only then set that sum equal to mv^2/r .

Mistake 3: Confusing centripetal acceleration with tangential acceleration

Centripetal acceleration ($a_c = v^2/r$) changes the direction of velocity. Tangential acceleration changes the speed. For uniform circular motion, only centripetal acceleration is present. For non-uniform circular motion (changing speed on a circular path), both are present.

Mistake 4: Using weight for centripetal force in orbital problems

In orbital problems, gravity IS the centripetal force: set $F_g = mv^2/r$. The satellite's weight is not zero; it equals F_g , which provides the centripetal acceleration. The sensation of weightlessness in orbit is not due to zero gravity but to free fall.

Mistake 5: Not converting units before using gravitational formulas

$G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ requires mass in kg and distance in meters. Convert all distances to meters and masses to kilograms before substituting.

Chapter Summary

Circular motion requires a net inward (centripetal) force. Gravity is the centripetal force for orbits. Newton's law of universal gravitation governs all gravitational interactions.

- Uniform circular motion: constant speed, continuously changing velocity direction; always accelerating.
 - Centripetal acceleration: $a_c = v^2/r$, directed toward center of circle.
 - Centripetal force: $F_c = mv^2/r$, directed toward center; provided by real forces (tension, friction, gravity, normal force).
 - Centripetal force is not a new force type — it is the label for whatever net inward force maintains circular motion.
 - Banking angle for a curve: $\tan\theta = v^2/(rg)$ for friction-free cornering.
 - Newton's law of gravitation: $F_g = G m_1 m_2 / r^2$; attractive; inverse-square law.
 - Gravitational field $g = GM/r^2$; decreases with altitude; near Earth's surface $\approx 9.81 \text{ m/s}^2$.
 - Orbital speed: $v = \sqrt{GM/r}$; depends only on M and r , not on satellite mass.
 - Kepler's Third Law: $T^2 = (4\pi^2/GM) r^3$.
 - Escape velocity: $v_{\text{esc}} = \sqrt{2GM/r} = \sqrt{2} \times v_{\text{orbit}}$.
 - Orbiting astronauts are weightless in sensation but not in gravity — they are in continuous free fall.
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Key Terms

uniform circular motion: Motion at constant speed along a circular path; velocity is always tangent to the circle; acceleration is always toward the center.

centripetal acceleration: The acceleration directed toward the center of a circular path: $a_c = v^2/r$; responsible for changing the direction of velocity.

centripetal force: The net inward force required to maintain circular motion: $F_c = mv^2/r$; not a new force type but a label for the net inward real force.

period (T): The time for one complete revolution; $T = 2\pi r/v$ for uniform circular motion.

frequency (f): The number of complete revolutions per second; $f = 1/T$; units: Hz.

centrifugal force: A fictitious outward force that appears in rotating reference frames; does not exist in inertial frames; should NOT be included in FBDs.

universal gravitation: Newton's law that every mass attracts every other mass: $F_g = Gm_1m_2/r^2$.

gravitational constant (G): The universal constant in Newton's gravity law: $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$.

gravitational field: The gravitational force per unit mass at a point in space; $g = GM/r^2$ directed toward M; units m/s^2 .

orbital speed: The speed required to maintain a circular orbit at radius r : $v = \sqrt{GM/r}$.

Kepler's Third Law: The orbital period squared is proportional to the orbital radius cubed: $T^2 = (4\pi^2/GM)r^3$.

escape velocity: The minimum speed to escape a gravitational field: $v_{\text{esc}} = \sqrt{2GM/r}$.

geosynchronous orbit: The orbital radius at which the period equals Earth's rotation period (24 h); $\approx 42,200$ km from Earth's center; satellite appears stationary from ground.

Concept Review Questions

1. Explain why an object moving at constant speed in a circle is still accelerating. In what direction is this acceleration?
2. What is centripetal force? Is it a new type of force? Give three physical examples of forces that can serve as centripetal force.
3. A car moves faster around a curve of the same radius. What happens to the required centripetal force? What happens if the radius doubles at the same speed?
4. Why do astronauts on the ISS feel weightless even though gravity is nearly as strong there as at Earth's surface?

5. How does Newton's law of universal gravitation differ from the simple $W = mg$ formula for weight? When is $W = mg$ a valid approximation?
 6. Explain why orbital speed decreases with increasing orbital radius. Use the formula $v = \sqrt{GM/r}$ in your explanation.
 7. What is Kepler's Third Law? Was it discovered before or after Newton's law of gravitation?
 8. Explain the concept of escape velocity in your own words. Why is it $\sqrt{2}$ times the orbital speed?
 9. Why is the centrifugal force not included in free-body diagrams for circular motion in inertial frames?
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Circular Motion Problems

10. A 0.40 kg ball on a 0.75 m string swings in a horizontal circle at 5.0 m/s. Find: (a) centripetal acceleration; (b) tension in the string.
 11. A car of mass 1100 kg rounds a flat curve ($r = 50.0$ m) at 15.0 m/s. (a) Find the centripetal force. (b) What is the minimum μ_s needed?
 12. A highway curve of radius 200 m is banked at 12° . At what speed can a car round it without relying on friction?
 13. A roller coaster loop has radius 15.0 m. (a) What minimum speed at the top keeps the car on the track? (b) What normal force does a 70.0 kg rider experience at the bottom if traveling at 22.0 m/s?
 14. A 1.5 kg mass swings on a 1.2 m rope in a vertical circle. (a) What is the tension at the bottom of the swing when moving at 4.0 m/s? (b) What is the tension at the top if moving at 3.5 m/s?
 15. A satellite travels in a circular orbit of radius 8.0×10^6 m around Earth. Find: (a) centripetal acceleration; (b) orbital speed; (c) period.
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Gravitation Problems

16. Find the gravitational force between Earth ($m = 5.97 \times 10^{24}$ kg) and the Moon ($m = 7.35 \times 10^{22}$ kg) separated by 3.84×10^8 m.

17. Two identical 5.0 kg balls are 0.50 m apart. Find the gravitational force between them.
 18. At what height above Earth's surface does $g = 4.905 \text{ m/s}^2$ (half of surface value)? (Earth radius = $6.37 \times 10^6 \text{ m}$)
 19. Compare the gravitational force on a 70 kg person at Earth's surface to the force at 400 km altitude.
 20. What is the weight of a 50.0 kg astronaut on the Moon ($g_{\text{Moon}} = 1.62 \text{ m/s}^2$)? On Mars ($g_{\text{Mars}} = 3.71 \text{ m/s}^2$)?
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Orbital Mechanics Exercises

21. Find the orbital speed and period of a satellite at $r = 7.0 \times 10^6 \text{ m}$ from Earth's center. ($M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$)
 22. The Moon orbits Earth in $T = 27.3$ days at $r = 3.84 \times 10^8 \text{ m}$. Verify this is consistent with $g = 9.81 \text{ m/s}^2$ at Earth's surface using the inverse-square law.
 23. Find the radius of geosynchronous orbit ($T = 86,400 \text{ s}$). ($M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$)
 24. Find the escape velocity from Earth's surface. From the Moon's surface ($M_{\text{Moon}} = 7.35 \times 10^{22} \text{ kg}$, $R_{\text{Moon}} = 1.74 \times 10^6 \text{ m}$).
 25. Planet X has twice Earth's mass and twice Earth's radius. Compare its surface g to Earth's. Compare escape velocity.
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Applied Physics Problems

26. A pilot pulls out of a dive at 180 m/s. The radius of curvature at the bottom is 1500 m. Find: (a) centripetal acceleration as a multiple of g ; (b) the normal force on a 75.0 kg pilot.
27. A 0.30 kg puck slides in a horizontal circle on a frictionless table, connected by a string through a hole to a hanging 0.20 kg mass. If the puck moves at 1.5 m/s, what is the radius of the circle?

28. (Challenge) A conical pendulum: a 0.50 kg ball on a 1.0 m string swings in a horizontal circle with the string at 30° from vertical. Find: (a) the radius of the circle; (b) the tension; (c) the speed.
29. Find the orbital period of a hypothetical satellite orbiting just above Earth's surface ($r = R_{\text{Earth}} = 6.37 \times 10^6 \text{ m}$). Compare your answer to $\sqrt{R/g}$.

Chapter 13 Checkpoint

I can...	Section
Explain why constant-speed circular motion requires acceleration	13.1
Calculate centripetal acceleration using $a_c = v^2/r$	13.3
Calculate centripetal force using $F_c = mv^2/r$	13.4
Identify the real force providing centripetal acceleration in various situations	13.5
Find the ideal banking angle for a given speed and radius	13.6
Apply Newton's law of gravitation $F_g = Gm_1m_2/r^2$	13.8
Calculate g at any distance from Earth using $g = GM/r^2$	13.9
Find orbital speed using $v = \sqrt{GM/r}$	13.11
Apply Kepler's Third Law $T^2 = (4\pi^2/GM)r^3$	13.12
Calculate escape velocity using $v = \sqrt{2GM/r}$	13.13
Explain weightlessness in orbit as free fall, not zero gravity	13.10
Avoid adding fictitious centrifugal force to FBDs	13.14

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